

GET 324 Mini Project Report

Concrete Bridge Deck Crack Detection

This project developed a deep learning application for detecting cracks on concrete bridge deck surfaces using the Deck subset of the SDNET2018 dataset. The trained EfficientNetV2-B0 model was deployed as a Streamlit web application, allowing users to upload bridge deck images and receive instant crack classification results. A major challenge encountered was poor model prediction caused by an imbalanced dataset, which was addressed by balancing the dataset before training. Another challenge was the large model size, making deployment more difficult. Future improvements include optimizing the model for faster deployment, improving cracked-class recall, and evaluating the model with additional datasets and cross-validation techniques.

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